EXPERIMENT 6

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Experiment 6 – Understanding and Implementing Cursors for Row-by-Row Data Processing (Oracle, Accenture, Scaler, D. E. Shaw & Co)

Experiment

Experiment 6: Working with PL/SQL cursors to process multiple rows of a table individually. This experiment demonstrates the use of implicit cursors, explicit cursors, and cursor attributes to implement business logic on row-by-row data in Oracle and PostgreSQL.

Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Objective

- To implement and analyze implicit cursors for single-row operations.
- To implement explicit cursors to fetch and process multiple records.
- To apply cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN) for controlling program flow.
- To process employee records row by row and apply business logic effectively.

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

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Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

Problem Statement

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules. Using cursors allows row-by-row processing for detailed business logic implementation.

Practical / Experiment Steps

1. Design PL/SQL programs that demonstrate:
    - The use of implicit cursors for single-row DML operations
    - The use of explicit cursors to fetch and process multiple records
    - The application of cursor attributes to control program execution
2. Create PL/SQL programs to:
    - Fetch employee records from a database table using cursors
    - Process each record individually
    - Display results or apply business logic using cursor attributes

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Procedure

1. Open Oracle SQL Developer or pgAdmin and connect to the database.
2. Create an employee table with sample data (id, name, department, salary).
3. Write an anonymous PL/SQL block demonstrating implicit cursor operations.
4. Write a PL/SQL block using an explicit cursor to fetch and process multiple records.
5. Apply cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN to control execution flow.
6. Execute the blocks and observe the output.

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7. Capture results and screenshots for documentation.

Input / Output Details

Input

employee table:

- id (Integer)
- name (Varchar)
- department (Varchar)
- salary (Integer)

Cursor logic used:

- Implicit cursor for single-row processing
- Explicit cursor for multi-row processing
- Cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN)

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Step-wise Output

Step 1 – Create employee Table

[SQL Worksheet]*

```
CREATE TABLE EMPLOYEE (
    EMP_ID NUMBER PRIMARY KEY,
    EMP_NAME VARCHAR2(50),
    SALARY NUMBER
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE TABLE EMPLOYEE (
         EMP_ID NUMBER PRIMARY KEY,
         EMP_NAME VARCHAR2(50),
         SALARY NUMBER...
Show more...


Table EMPLOYEE created.

Elapsed: 00:00:00.019
```

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Step 2 – Insert Data into employee Table

[SQL Worksheet]*

```sql
INSERT INTO EMPLOYEE VALUES (101, 'Amit', 25000);
INSERT INTO EMPLOYEE VALUES (102, 'Riya', 32000);
INSERT INTO EMPLOYEE VALUES (103, 'Rahul', 45000);
INSERT INTO EMPLOYEE VALUES (104, 'Sneha', 28000);
INSERT INTO EMPLOYEE VALUES (105, 'Arjun', 50000);


COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL histor

```
SQL> INSERT INTO EMPLOYEE VALUES (101, 'Amit', 25000)



1 row inserted.

Elapsed: 00:00:00.021
```

```
SQL> INSERT INTO EMPLOYEE VALUES (102, 'Riya', 32000)



1 row inserted.

Elapsed: 00:00:00.002
```

Step 3 – Implicit Cursor Operation

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[SQL Worksheet]*

```
BEGIN
    UPDATE EMPLOYEE
    SET SALARY = SALARY * 1.10
    WHERE SALARY < 30000;

    DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || SQL%ROWCOUNT);

    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Update Successful');
    END IF;
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> BEGIN
         UPDATE EMPLOYEE
         SET SALARY = SALARY * 1.10
         WHERE SALARY < 30000;...
Show more...


Rows Updated: 1
Update Successful


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.098
```

Step 4 – Explicit Cursor Declaration and Fetch

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[SQL Worksheet]*

```
DECLARE
    CURSOR emp_cursor IS
        SELECT EMP_ID, EMP_NAME, SALARY FROM EMPLOYEE
        WHERE SALARY > 30000;

    v_id EMPLOYEE.EMP_ID%TYPE;
    v_name EMPLOYEE.EMP_NAME%TYPE;
    v_salary EMPLOYEE.SALARY%TYPE;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_salary;
        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'ID: ' || v_id ||
            ' Name: ' || v_name ||
            ' Salary: ' || v_salary
        );
    END LOOP;

    CLOSE emp_cursor;
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
WHERE SALARY > 30000,...
Show more...


ID: 101 Name: Amit Salary: 30250
ID: 102 Name: Riya Salary: 32000
ID: 103 Name: Rahul Salary: 45000
ID: 104 Name: Sneha Salary: 30800
ID: 105 Name: Arjun Salary: 50000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.013
```

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Step 5 – Process Records Using Cursir Attributes

[SQL Worksheet]*

```
DECLARE
    CURSOR emp_cursor IS SELECT * FROM EMPLOYEE;
    emp_record EMPLOYEE%ROWTYPE;

BEGIN
    OPEN emp_cursor;

    IF emp_cursor%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor Opened Successfully');
    END IF;

    LOOP
        FETCH emp_cursor INTO emp_record;
        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'Employee: ' || emp_record.EMP_NAME ||
            ' | Salary: ' || emp_record.SALARY
        );
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Total Rows Processed: ' || emp_cursor%ROWCOUNT

    CLOSE emp_cursor;
END;
/
```

Query result | **Script output** | DBMS output | Explain Plan | SQL history

```
Cursor Opened Successfully
Employee: Amit | Salary: 30250
Employee: Riya | Salary: 32000
Employee: Rahul | Salary: 45000
Employee: Sneha | Salary: 30800
Employee: Arjun | Salary: 50000
Total Rows Processed: 5


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.013
```

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Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the role of cursors in PL/SQL for handling multi-row query results.
- Differentiate between implicit cursors and explicit cursors.
- Use cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN.
- Develop PL/SQL programs that process database records row by row.
- Apply cursor-based logic in real-world business scenarios used in organizations like Oracle, Accenture, Scaler, and D. E. Shaw & Co.